**1.What is Chi-Square Test?**

**Definition**

The **Chi-Square Test** is a statistical test used to determine whether there is a **significant difference between observed and expected frequencies**.

It is mainly used for **categorical data** (non-numerical data like gender, color, category, etc.).

**Formula**

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where:

- **O**= Observed frequency
- **E** = Expected frequency

**Types of Chi-Square Test**

**1. Chi-Square Goodness of Fit Test**

**Definition**

This test checks whether the **observed data fits a particular expected distribution**.

**Example**

- Checking if a dice is fair
- Expected: each number appears equally
- Compare with actual results

**Use Case**

- Testing randomness
- Checking distribution fit

**2. Chi-Square Test of Independence**

**Definition**

This test checks whether **two categorical variables are independent or related**.

**Example**

- Gender vs Product preference
- Are they related or independent?

**Use Case**

- Market research
- Surveys
- Association between variables

**2. How does SVD help in dimensionality reduction?**

**Definition**

**Singular Value Decomposition (SVD)** is a matrix factorization technique that decomposes a data matrix into three matrices:

$$A = U\Sigma V^T$$

Where:

- **U** → Left singular vectors
- **Σ (Sigma)** → Diagonal matrix of singular values
- **$V^T$** → Right singular vectors

**How SVD reduces dimensionality**

1. **Captures important information**
    - Singular values in **Σ** represent the importance (variance) of features.
2. **Ranks features by importance**
    - Larger singular values → more important patterns
    - Smaller values → noise or less useful information
3. **Truncation (key step)**
    - Keep only top **k largest singular values**
    - Remove the rest (low-impact components)
4. **Reduced representation**
    - Reduces number of dimensions from **n → k (k < n)**
5. **Noise reduction**
    - Eliminating small singular values removes noise
6. **Data compression**
    - Stores less data while preserving structure
7. **Better performance**
    - Speeds up machine learning algorithms

**3. What is the Curse of Dimensionality & how to overcome it?**

**Definition**

The **Curse of Dimensionality** refers to problems that arise when the number of features (dimensions) increases in a dataset.

**Problems caused**

1. **Data sparsity**
    - Data points become widely spread
2. **Distance becomes meaningless**
    - Hard to measure similarity
3. **Increased computation cost**
    - More memory and time required
4. **Overfitting**
    - Model learns noise instead of patterns
5. **Need for more data**
    - High dimensions require huge datasets

**How to overcome / reduce it**

1. **Dimensionality Reduction Techniques**
    - Principal Component Analysis (PCA)
    - Singular Value Decomposition (SVD)
2. **Feature Selection**
    - Keep only important features
    - Remove irrelevant or redundant ones
3. **Regularization**
    - Techniques like L1/L2 reduce overfitting
4. **Increase training data**
    - More data helps cover space better
5. **Use simpler models**
    - Avoid overly complex models
6. **Feature engineering**
    - Combine or transform features

**4. What is PCA and does it help in reducing dimensionality?**

**Definition**

**Principal Component Analysis (PCA)** is a statistical technique used to transform data into a new set of variables called **principal components**.

**How PCA reduces dimensionality**

1. **Selects most important components**
    - Keeps components with highest variance
2. **Removes redundancy**
    - Eliminates correlated features
3. **Projects data into lower dimension**
    - From n features → k components (k < n)
4. **Preserves maximum information**
    - Retains most variance of original data

**5.What is CSV?**

**Definition**

**CSV (Comma-Separated Values)** is a simple file format used to store tabular data (like a table or spreadsheet).

- Each **row** represents a record
- Each **value is separated by a comma (,)**
- It is a **plain text file**

**Example of CSV File**

Name,Age,Marks

Ravi,20,85

Anita,21,90

Rahul,19,78

**Features**

- Easy to read and write
- Supported by tools like Excel, databases, and R
- Lightweight and portable

**How to Load a CSV File in R?**

**Basic Syntax**

```
data <- read.csv("file_name.csv")
```

**Example**

```
data <- read.csv("students.csv")
print(data)
```

**From a Specific Path**

```
data <- read.csv("C:/Users/YourName/Documents/students.csv")
data <- read.csv("students.csv")
```

**Methods / Functions to Read CSV Files in R**

**1. read.csv()**

**Description**

- Most commonly used function
- Reads CSV file with comma separator

```
data <- read.csv("file.csv")
```

**2. read.table()**

**Description**

- More general function
- Can read different formats

```
data <- read.table("file.csv", sep = ",", header = TRUE)
```

**3. read.csv2()**

**Description**

- Used when separator is **semicolon (;)**
- Common in European formats

```
data <- read.csv2("file.csv")
```

**4. read.delim()**

**Description**

- Used for **tab-separated files**

```
data <- read.delim("file.txt")
```